A TAUT CONTINUOUS MAP WITH DENSE IMAGE OUT OF A COMPLETE FIRST COUNTABLE RING IS SURJECTIVE AND OPEN

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof span cites one cross-chapter lemma and one same-chapter lemma; both are delivered as navigation stubs. No bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The open mapping lemma for complete linearly topologized rings with a countable fundamental system of open ideals. More on Algebra, Lemma more-algebra-lemma-open-mapping, invoked at proof lines 1469-1471. Yields openness of the map, combined with tautness of the homomorphism.
- Quotients of complete linearly topologized rings by closed ideals. Same chapter, Lemma lemma-quotient-by-closed, invoked at proof lines 1475-1476. Gives completeness of B, identified with A/K for the closed kernel K.
- Tautness of a continuous homomorphism of linearly topologized rings. Same chapter, Definition definition-taut. The hypothesis that is combined with the open mapping lemma.

Source reference. Original lemma 0AMT: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 972–972. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0CQW: sources/stacks-chow/tags.txt, source lines 15976–15976. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $\varphi : A \to B$ be a continuous homomorphism of linearly topologized rings. Assume*

(1) *φ is taut and has dense image,*
(2) *A is complete and has a countable fundamental system of open ideals, and*
(3) *B is separated.*

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Then φ is surjective and open, B is complete, and $B = A/K$ for some closed ideal $K \subset A$.

Proof. By the open mapping lemma (More on Algebra, Lemma 0CQW) combined with tautness of φ, we see the map φ is open. Since the image of φ is dense, we see that φ is surjective. The kernel K of φ is closed as φ is continuous. It follows that $B = A/K$ is complete, see for example Lemma 0AMT. $\square$